

STRATEGY API CONTRACT & ADMIN TRACING DOCUMENTATION

Version: 2.0.0 | Execution Engine: Event-Driven Quantitative Engine | Default Mode: PAPER

Executive Overview for Frontend Developers & Administrators:

This API enables the full lifecycle of custom trading strategies. A single normalized JSON schema (Schema v2.0.0) powers all strategy archetypes: single-leg option buying (EMA Pullback, RSI Alert Candle) and 4-leg hedged option selling (Camarilla S3/R3 Pivot Reversal). **Pipeline:** Frontend JSON → Create (Dual Persistence) → Activate (Runtime State Machine) → Market Event → Strategy Engine → Golden Rules → Signal Engine → Risk Approval → Execution Validator → Execution Engine (PAPER/Mock) → Position Open → Exit Engine → Position Closed.

1. Strategy Lifecycle & State Machine Architecture

Every strategy operates strictly under a deterministic state machine managed by StrategyStateManager. The state controls what market events or exit checks the strategy is eligible to process.

Lifecycle State	State Description	Allowed Next Transitions
WAITING	Initial state after creation. Idle and un-activated.	ELIGIBLE, ARCHIVED
ELIGIBLE	Validated and ready for scheduling / activation.	MONITORING_ENTRY, PAUSED, WAITING
MONITORING_ENTRY	Active in Market Router. Receiving live candle/tick events to evaluate entry rules.	ENTRY_SIGNAL, PAUSED, ELIGIBLE
ENTRY_SIGNAL	Entry condition matched & Golden Rules approved. Signal emitted to Risk Engine.	ORDER_PENDING, MONITORING_ENTRY
ORDER_PENDING	Risk approved. Order dispatched to Execution Engine / Paper Broker.	POSITION_OPEN, MONITORING_ENTRY
POSITION_OPEN	Paper order filled. User position opened and tracked in database.	MONITORING_EXIT, EXIT_ORDER_PENDING
MONITORING_EXIT	ExitEngine actively evaluating targets, stop loss, trailing stops, or 15:15 square-off.	EXIT_SIGNAL, EXIT_ORDER_PENDING
EXIT_SIGNAL	Exit condition triggered. EXIT signal emitted to Execution Engine.	EXIT_ORDER_PENDING, POSITION_OPEN
EXIT_ORDER_PENDING	Exit square-off order executing at broker.	POSITION_CLOSED, POSITION_OPEN
POSITION_CLOSED	All legs closed and reconciled. Returns to MONITORING_ENTRY if re-entry allowed.	MONITORING_ENTRY, ELIGIBLE, WAITING
PAUSED / ARCHIVED	Manually paused by user or archived. Detached from market data router.	ELIGIBLE, WAITING, MONITORING_ENTRY

2. Endpoint Reference: Strategy Management

POST

`/api/strategy` (*Canonical*) or `/api/v1/strategies`

Why We Use This API: This is the primary entry point for frontend strategy creation. It accepts Schema Version 2.0.0 JSON payloads, performs pre-flight validation, creates the Strategy root entity, creates an immutable StrategyVersion (v1), persists relational legs/settings, serializes all rich builder fields into parameters JSON, and initializes the StrategyRuntimeState in WAITING state.

Authentication & Headers:

Authorization: Bearer <JWT_TOKEN> | Content-Type: application/json

Key Request Fields (Schema v2.0.0):

Field	Type	Required	Description & Notes
schemaVersion	string	Yes	Must be "2.0.0".
meta	object	Yes	{ strategyId, strategyName, authorName, status }.
tradingHorizon	string	Yes	"Intraday", "Monthly", or "Positional".
timeframe	string	Yes	"5m", "15m", "1h", "Daily" / "1d".
instrument	object	Yes	{ underlying: "NIFTY 50" "RELIANCE", expiryType: "Weekly" "Monthly" }.
schedule	object	Yes	{ entryFrom: "09:30", forcedExitTime: "15:15", applicableDays }.
entryConditions	array	Yes	List of indicator / price / candlestick rules (e.g. EMA 8 > EMA 33, RSI > 60).
exitConditions	array	Yes	List of exit rules (Target 2R, SL hit, opposite crossover, 15:15 forced exit).
riskManagement	object	Yes	{ maxLossPerTrade, maxLossPerDay, maxLossPerWeek, maxTradesPerDay, stopLoss }.
target	object	Yes	{ value: "2R" "Alert Candle Range" "2.5%", targets }.
options	object	Yes	{ strikeSelection: "ATM", legs: [...] }. Contains 4 legs for multi-leg.
execution	object	Yes	{ orderType: "MARKET" "LIMIT", slippage: 0.10, orderTimeout: 30 }.
mode	string	No	Defaults to "PAPER". Live mode protected.

cURL Example:

```
curl -X POST "https://api.zenalgo.com/api/strategy" \
-H "Authorization: Bearer YOUR_JWT_TOKEN" \
-H "Content-Type: application/json" \
-d '{
  "schemaVersion": "2.0.0",
  "meta": {"strategyId": "EMA_8_33_PULLBACK", "strategyName": "Nifty EMA 8/33 Scalper"},
  "tradingHorizon": "Intraday",
  "timeframe": "5m",
  "instrument": {"underlying": "NIFTY 50", "expiryType": "Weekly"},
  "schedule": {"entryFrom": "09:30", "forcedExitTime": "15:15"},
  "entryConditions": ["8 EMA crosses above 33 EMA", "Pullback to 8 EMA"],
  "riskManagement": {"maxLossPerTrade": 2500.0, "riskRewardRatio": "1:2"},
  "target": {"value": "2R"},
  "options": {"strikeSelection": "ATM"},
  "execution": {"orderType": "MARKET"}
}'
```

Response Example (201 Created):

```
{
  "success": true,
  "message": "Strategy created successfully",
  "data": {
    "id": 1042,
    "currentVersionId": 2085,
    "name": "Nifty EMA 8/33 Scalper",
    "status": "DRAFT",
    "mode": "PAPER",
    "timeframe": "5m",
    "underlying": "NIFTY",
    "capital": "100000.00",
    "legs": [{"sequence": 1, "segment": "OPT", "side": "BUY", "strikeSelection": "ATM"}],
    "schedule": {"entryFrom": "09:30", "forcedExitTime": "15:15"}
  }
}
```

GET

/api/strategy/{id} or /api/v1/strategies/{id}

Why We Use This API: Used by the frontend builder to reload the full strategy configuration when viewing, editing, or analyzing backtest/paper results. It deserializes and returns all original builder fields (including 4-leg option definitions, pivot configuration, earnings exclusion, risk limits, and trailing stops) exactly as sent during creation.

cURL Example:

```
curl -X GET "https://api.zenalgo.com/api/strategy/1042" -H "Authorization: Bearer YOUR_JWT_TOKEN"
```

POST

/api/strategy/{id}/activate or /api/v1/strategies/{id}/activate

Why We Use This API: Activates a strategy for autonomous paper trading. It validates strategy completeness, transitions runtime state from WAITING → ELIGIBLE → MONITORING_ENTRY, and dynamically registers the strategy with StrategyRouter to start processing live market tick/candle events. **Idempotent:** Activating an already active strategy is safe and returns the active strategy state.

cURL Example:

```
curl -X POST "https://api.zenalgo.com/api/strategy/1042/activate" -H "Authorization: Bearer YOUR_JWT_TOKEN"
```

3. Admin Observability & End-to-End Tracing Architecture

For compliance, debugging, copy-trading management, and financial auditability, the system records a complete, immutable correlation chain across all domain layers. Administrators can trace every order and fill back to the exact candle, market event key, strategy version, and risk approval trace.

Tracing Field	Originating Domain	Format / Example	Audit Purpose
strategy_id	Strategy Domain	1042	Identifies the user's root strategy entity.
strategy_version_id	Strategy Domain	2085	Guarantees immutability. Trades execute on the exact version configured.
market_event_key	Market Data	1042:NIFTY:5m:2026-09-01T09:35:00	Ensures exactly-once processing per candle/tick event.
signal_key	Signal Engine	SIG-ENTRY-1042-2085-20260901-0935	Deterministic signal key preventing duplicate trade triggers.
execution_batch_id	Signal Domain	5501	Groups all copy-trading users associated with a signal.
correlation_id	Risk & Execution	TRACE-5501-1042-USR-88-a1b2c3d4	Unique end-to-end trace per user execution.
execution_id	Execution Engine	7742	Persisted execution record linking to filled broker orders.
position_id	Position Tracker	9903	Active open position with real-time net quantity & PnL.
exit_signal_key	Exit Engine	SIG-EXIT-1042-9903-TARGET-2R	Exit condition record (Target 2R, Stop Loss, or 15:15 square-off).

4. Admin Strategy & Execution Tracking Endpoints

HTTP Method & Route	Role Required	Purpose & Functionality
GET /api/v1/admin/strategies	SUPER_ADMIN, ADMIN	Query all system strategies with user filter, status filter, and pagination.
GET /api/v1/admin/strategies/{id}	SUPER_ADMIN, ADMIN	Fetch detailed strategy configuration, execution logs, and runtime state.
POST /api/v1/admin/strategies	SUPER_ADMIN, ADMIN	Admin-level strategy creation and provisioning on behalf of users.
POST /api/v1/admin/strategies/{id}/activate-paper	SUPER_ADMIN, ADMIN	Admin override to activate any user strategy directly into PAPER trading.
POST /api/v1/admin/strategies/{id}/activate-live	SUPER_ADMIN	Protected live activation switch. Requires environment production flag and broker approval.
GET /api/v1/admin/execution/strategies/{id}/batches	SUPER_ADMIN, ADMIN	Track how many users executed: Total subscribers, eligible users, successful fills, failures.
GET /api/v1/admin/execution/batches/{id}/traces	SUPER_ADMIN, ADMIN	Track all users in batch: Per-user status (EXECUTED, FAILED, REJECTED), failure code & reason.
GET /api/v1/admin/execution/batches/{id}/failures	SUPER_ADMIN, ADMIN	Aggregated failure breakdown: Grouped counts (e.g. BROKER_SESSION_INVALID: 4, INSUFFICIENT_FUNDS: 2).
GET /api/v1/admin/execution/strategies/{id}/placed-orders	SUPER_ADMIN, ADMIN	Placed Paper Orders & Positions: Lists executed paper trades, filled legs, execution prices, and live PnL.

5. Supported Reference Strategy Archetypes (Payload Matrix)

Strategy Name	Archetype	Underlying / TF	Leg Structure & Execution	Exit Rules
Strategy 1 EMA_8_33_PULLBACK	Trend Following Option Buying	NIFTY 50 5m timeframe Weekly Expiry	Single-leg dynamically resolved ATM BUY CE / PE . Market order, 0.10% slippage.	Target = 2R. SL = Confirmation candle low/high. 50% partial exit at 1R. Forced exit = 15:15.
Strategy 2 RSI_60_ALERT_CANDLE	Momentum Breakout Option Buying	NIFTY 50 15m timeframe Weekly Expiry	Single-leg ATM BUY CE on alert candle high breach / BUY PE on low breach.	Target = Alert Candle Range. SL = Alert candle low/high. Trailing stop enabled. Forced exit = 15:15.
Strategy 3 RELIANCE_S3_R3_PIVOT_REVERSAL	Mean Reversion Option Selling (Hedged)	RELIANCE Daily timeframe Monthly Expiry	ALL 4 LEGS PRESERVED: • Leg 1: SELL CE (ATM) • Leg 2: BUY CE (+200pt Hedge) • Leg 3: SELL PE (ATM) • Leg 4: BUY PE (-200pt Hedge)	Target = 2.5% Premium Decay. SL = R3/S3 breach. Quarterly earnings excluded. Simultaneous 4-leg square-off.

6. Standard Error Codes & Validation Rules

Error Code	HTTP Status	Root Cause	Resolution / Action
TIME_REQUIRED	400	Schedule time format is missing or invalid.	Provide time in HH:MM format (e.g. "09:30").
INVALID_UNDERLYING	400	Underlying not in supported list.	Use NIFTY, BANKNIFTY, FINNIFTY, RELIANCE, TCS, etc.
INVALID_EXECUTION_PACKAGE	422	Short option leg missing mandatory hedge leg.	Ensure multi-leg option selling includes protective BUY leg.
EXECUTION_DUPLICATE	409	Execution request already processed for signal.	Ignored safely by idempotency boundary.
RISK_LIMIT_EXCEEDED	422	Daily trade count or max loss threshold exceeded.	Strategy enters cooldown or halts for the trading session.